

Digital Forensics Laboratory Report

Module 7: Data Carving and File Recovery Techniques

Student Name: Idriss Muhammad Abdullahi

Registration Number: fwsd2511475

Course: Cybersecurity and Digital Forensics

Lab Title: Data Carving and File Recovery Techniques

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1. Introduction

This laboratory exercise focused on the practical application of data carving and file recovery techniques used in digital forensic investigations. The aim was to understand how files can be identified, reconstructed, and recovered using file signatures, hexadecimal analysis, forensic tools, and cryptographic hash verification.

The laboratory activities included examining JPEG file signatures, reconstructing files from hexadecimal data, analysing metadata-like information, discovering embedded content using Binwalk, preparing a forensic USB image, and recovering deleted files using Scalpel. All evidence was handled in a forensically sound manner without modifying the original files.

2. Evidence Acquisition and Integrity Verification

A dedicated working directory was created to store all evidence files.

The evidence files provided for this laboratory were:

- Ch01InChap01.dd
- J_ub_law.jpg
- 120M.7z

Before beginning the analysis, cryptographic hashes were calculated for each file using MD5 and SHA-256 hashing algorithms. This process ensured that the integrity of the evidence could be verified throughout the investigation.

Commands used:

- md5sum Ch01InChap01.dd
- sha256sum Ch01InChap01.dd

The generated hash values were recorded and preserved as part of the investigation documentation.

3. Part A - JPEG File Signature and Hexadecimal Analysis

The JPEG sample file, **J_ub_law.jpg**, was analysed using the xxd utility to inspect its hexadecimal content.

```
xxd J_ub_law.jpg | head
```

The beginning of the file contained the standard JPEG Start of Image (SOI) signature:

```
FF D8
```

Further inspection showed the JFIF header, confirming that the file was a valid JPEG image.

The end of the file was examined using:

```
xxd J_ub_law.jpg | tail -n 2
```

The JPEG signature was identified as:

```
FF D9
```

These signatures confirmed the file format and demonstrated the importance of file headers and footers during data carving investigations.

Creating and Reconstructing a Hex Dump

A hexadecimal dump of the image was created using:

```
xxd J_ub_law.jpg > J_ub_law.hex
```

The image was then reconstructed from the hexadecimal dump using:

```
xxd -r J_ub_law.hex reconstructed.jpg
```

Hash Verification

Both the original image and the reconstructed image were hashed using MD5 and SHA-256.

The hash values matched exactly, proving that the reconstructed image was identical to the original file and had not been altered during the process.

4. Part B - Metadata and String Analysis

The JPEG image was further examined to identify any visible metadata-like information.

Linux command used:

```
strings J_ub_law.jpg
```

The output was reviewed for any available information such as:

- Camera details
- Software information
- Dates

Any information discovered was documented exactly as displayed by the tool.

5. Part D - Embedded File Discovery Using Binwalk

Binwalk was used to identify embedded file signatures and hidden content within the supplied evidence files.

L commands used:

binwalk J_ub_law.jpg

binwalk Ch01InChap01.dd

Automatic extraction was not performed as demonstrated in the lecture note

6. Part E - USB Image Preparation

The archived evidence file was extracted using the 7-Zip utility.

7z x 120M.7z

After extraction, the resulting forensic image file was identified and preserved.

Hash values were calculated using: hashdeep -c md5, sha1 usb_fat_carving.001

The integrity of the image was verified before further analysis.

Additional examination was carried out using forensic tools to identify image characteristics and confirm the presence of a recoverable file system structure.

7. Part F - File Recovery Using Scalpel

Scalpel was configured to recover JPEG image files from the forensic USB image.

The configuration file was modified to enable JPEG recovery signatures.

The carving operation was started using:

scalpel imagefile -o scalpel_output as demonstrate in module 7 materials

Scalpel successfully completed the recovery process and created an output directory containing recovered files and an audit log.

Audit Log Verification

The file: audit.txt

was reviewed to verify the carving process and document recovered artefacts.

Hash Verification

Recovered files were hashed using MD5 and SHA-256 to support evidence validation and documentation.

8. Findings

The forensic examination produced the following results:

- JPEG file signatures were successfully identified.

- The JPEG image was converted into a hexadecimal dump.
- The image was successfully reconstructed from hexadecimal data.
- Hash verification confirmed file integrity.
- Metadata-like information was examined using strings.
- Embedded content analysis was successfully performed with Binwalk.
- The forensic USB image was extracted and verified.
- Scalpel successfully recovered deleted JPEG files.
- Recovered images opened successfully and were verified using cryptographic hashes.

9. Conclusion

This laboratory successfully demonstrated the principles of data carving and file recovery within a digital forensic investigation. File signature analysis, hexadecimal reconstruction, metadata examination, embedded file discovery, USB image preparation, and file carving techniques were successfully completed.